

Question	Answer	Marks
2(a)	product of force and distance	M1
	perpendicular distance between the (line of action of the two) forces	A1
2(b)	straight vertical line drawn from centre to bottom corner	B1
2(c)(i)	centre of gravity is to the right of the rod so torque is anticlockwise or moment of weight of sheet about rod is clockwise so torque is anticlockwise	B1
2(c)(ii)	(horizontal displacement) = $3.3 / (2.8 \times 9.81) = 0.12$ (m)	A1
2(d)	$\rho = m / V$	C1
	$V = 4.0 \times 10^{-3} \times (\text{side length})^2$	C1
	side length = $\sqrt{2.8 / (3000 \times 0.0040)} = 0.48\text{m}$	A1
2(e)	point lies on a straight line at 45° to the horizontal from the bottom-right corner to the edge of the rod	B1
	point lies on a vertical line half-way between rod and right-hand edge	B1

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